Response to Reviewers

Editor comments

The manuscript presents a useful applied comparison of Bayesian and frequentist meta-analysis, but several claims about the relative merits of the two frameworks are overstated or lack sufficient support. The authors should refocus the paper as a comparison of two specific implementations rather than broad frameworks, clarify technical inaccuracies , and provide supporting analyses for statements about robustness to outliers and priors. Both reviewers have provided highly constructive and meaningful suggestions that should be carefully addressed by the authors. - Rongji Mu

Response to the Editor

We would like to thank the editor and reviewers for their time and careful reading of our manuscript and their valuable feedback. Below we respond to each point in turn; revised manuscript text is quoted verbatim.

Reviewer 1

This manuscript presents a systematic meta-analysis of the association between lead exposure and children's IQ and compares frequentist and Bayesian meta-analytic frameworks. The study is well organized and the methodology is clearly described. However, some statements comparing Bayesian and frequentist approaches appear overly strong or technically imprecise. A broad comparison between Bayesian and frequentist philosophies involves complex statistical issues that require extensive methodological expertise. This manuscript would be stronger if it focused on clearly presenting how the two approaches differ in this specific meta-analysis and how these differences affect the interpretation of the results, rather than making general claims about the superiority or limitations of one framework over the other. The following comments may help improve the manuscript.

Major comments

Reviewer 1, comment 1

In the Abstract, the authors state: "Sensitivity analyses showed robustness to RoB and prior specification." This statement may be misleading. For the prior sensitivity analysis, the results only indicate that, in this specific dataset, the likelihood information dominates the influence of the selected priors. This should not be interpreted as a general property of Bayesian analysis. Prior specification can be important, especially in small meta-analyses with limited evidence. In addition, the statement regarding robustness to RoB requires clarification. According to Table 5, the frequentist estimates appear at least similarly robust after excluding studies with high or moderate RoB. Therefore, the Abstract should avoid implying a particular advantage of the Bayesian framework.

Response

Thank the reviewer for this thoughtful comment. We agree and have revised the abstract to clarify that both frameworks were similarly robust to exclusion of studies with high or moderate RoB, and that Bayesian estimates were insensitive to the alternative priors examined in our sensitivity analysis — which is not a general property of Bayesian analysis.

Revised: Abstract – Results

"Sensitivity analyses showed both meta-analyses were robust to RoB, and Bayesian estimates were insensitive to tested priors."

Reviewer 1, comment 2

The manuscript states: "The Bayesian approach was less sensitive to outlier studies reporting substantial uncertainty." However, no specific analysis is provided to support this statement. The authors should clarify which studies are considered outliers and quantify how much the pooled estimates change after excluding these studies.

Response

We thank the reviewer for this comment and acknowledge that we mixed the two separate concepts of "outlier studies" and "studies with uncertain estimates". With the statement quoted above, we intended to discuss only how "studies with uncertain estimates" are affected by shrinkage.

To address the reviewers comment, we have therefore removed all references to "outlier studies" and replaced this with "studies reporting uncertain estimates" or comparable wording. Furthermore, we have expanded our sensitivity analysis to include prior settings that systematically vary the priors for either the location of μ or the scale of τ while holding the remaining settings constant:

While τ remained at its default, we varied μ:

- skeptical prior: $\mathcal{N}(-0.33, 2)$
- enthusiastic prior: $\mathcal{N}(-3, 2)$

While μ was set to a flat prior, we varied τ:

- strong shrinkage: $\mathcal{N}^+(0, 0.66)$
- weak shrinkage: $\mathcal{N}^+(0, 6)$

We set μ to a flat prior to see if we would recover the frequentist estimate for μ under weak shrinkage. With this baseline, the additional sensitivity analyses allow us to better highlight the different sensitivity of Bayesian and frequentist approaches to studies with uncertain estimates.

We believe that this analysis and the related changes significantly improve our manuscript, and would like to thank the reviewer once again for stimulating this with their comment.

Revised: Methods – Sensitivity Analysis

"To be able to identify the individual impact of the prior settings for μ and τ, we varied one of the two parameters while holding the other constant. The location of the μ prior was varied to values a third and three times that of the default value, while τ was held at $\mathcal{N}^+(0, 2)$, and the same was done for the scale of the τ prior while holding μ at $\mathcal{N}(0, 100)$ (Table 4). We decided to hold μ at $\mathcal{N}(0, 100)$ instead of using the values from the main analysis $\mathcal{N}(-1, 2)$ because it allowed us to examine if Bayesian and frequentist estimates converge under a very wide μ prior and weak shrinkage."

Revised: Results – Sensitivity Analysis – Prior Sensitivity

"Varying the prior on the location of μ while holding τ at $\mathcal{N}^+(0, 2)$, we found neither skeptical and enthusiastic priors on the location of μ substantially alter our conclusions. Varying the strength of shrinkage by decreasing or increasing the scale of τ while holding μ at $\mathcal{N}(0, 100)$ also did not substantially alter what we concluded from the estimates. The estimate for μ under weak shrinkage matched that of the frequentist analysis, while the estimate under strong shrinkage was notably more conservative. This is because many of the strongly negative coefficients were reported by studies with large uncertainty which are in turn more affected by shrinkage, pulling their estimands closer towards zero."

Revised: Discussion

"The two frameworks may differ in how they treat studies reporting substantial uncertainty depending on prior configuration and sample size. Bayesian meta-analysis can be set up such that it is less sensitive to these uncertain studies: a narrower prior distribution for the heterogeneity leads to stronger shrinkage, giving less weight to uncertain studies (Table 4). In contrast, frequentist models rely on inverse-variance weighting that is fully determined by the data – studies with large uncertainty are weighed less, but the degree to which this happens cannot be configured by the researcher. Setting the model up for stronger shrinkage via priors can therefore be particularly relevant in meta-analyses analysing only a small set of studies. Here, the observed heterogeneity may be high simply due to the small sample size, leaving our pooled estimate prone to being disproportionately impacted by a handful of uncertain studies."

Reviewer 1, comment 3

In the Background, the authors state: "Unlike frequentist confidence intervals, which are often misinterpreted as probabilistic..." This statement requires clarification. The issue is not that frequentist confidence intervals lack a probabilistic interpretation. Under the frequentist framework, confidence intervals are random intervals generated by a repeated sampling procedure and have a specified coverage probability. The common misunderstanding is that, after observing the data, the calculated confidence interval can be interpreted as having a 95% probability of containing the fixed but unknown parameter. Therefore, the distinction between frequentist confidence intervals and Bayesian credible intervals should be explained more precisely. The current wording may unintentionally suggest that confidence intervals are non-probabilistic, which is not accurate.

Response
We thank the reviewer for this comment, we changed the wording of this sentence to explain the difference between frequentist confidence intervals and Bayesian credible intervals more precisely.

Revised: Background
"Unlike frequentist confidence intervals, which are often misinterpreted as having a 95% probability of containing the unknown parameter [...]"

Reviewer 1, comment 4
In Section 3.3, the manuscript reports that the Bayesian model gives a larger between-study variance (tau^2) but a slightly smaller I^2 compared with the frequentist model. The authors state in Section 2.4.1 that I^2 was calculated using the typical within-study variance from the frequentist framework. However, the exact calculation of Bayesian I^2 is unclear. Please provide the formula or detailed explanation for how Bayesian I^2 was obtained and explain why a larger estimate of tau^2 resulted in a smaller I^2.

Response
We thank the reviewer for this comment. It helped us spot an inconsistency in how we summarised the posterior for τ^2.

We previously used the mean to summarise the right-skewed distribution of τ^2, giving stronger weight to very large samples of τ^2. This has lead to the apparent asymmetry between τ^2 and I^2 summaries that we thank the reviewer for pointing out. We carried this asymmetry forward by calculating I^2 for each sample *first* and *then* summarising as opposed to calculating a single I^2 from the mean of τ^2. An I^2 calculated this way would have been consistent with the reviewers intuition, resulting in an estimated I^2 of 86.6%, larger than the frequentist I^2 of 85.0%.

While providing consistent results, this approach would have discarded the uncertainty posterior distribution. We instead opted to summarise the posterior of τ^2 using the median, not the mean, as it more faithfully represents the typical value in a skewed distribution.

Using the median, the Bayesian estimates are now τ^2 = 3.3 (95% CrI: 1.0 to 9.5) and I^2 = 84.7% (95% CrI: 63.9% to 94.2%), against a frequentist τ^2 = 3.3 and I^2 = 85.0%. The apparent asymmetry is resolved, and the two frameworks now agree on heterogeneity, too.

We have updated the Abstract, Section 2.4.1, Section 3.3 and rewritten Additional File 4 accordingly.

Revised: Abstract – Results
"Between-study heterogeneity was substantial (frequentist: τ^2 = 3.3, I^2 = 85.0%; Bayesian: posterior median τ^2 = 3.3, I^2 = 84.7%)."

Revised: Methods – Bayesian Meta-Analysis
"Because the posterior distribution of τ^2 is right-skewed, we summarise τ^2 and I^2 by their posterior medians rather than their posterior means, as the median represents the typical value more faithfully and keeps the reported point estimates consistent with their credible intervals (for further detail, see additional file 4)."

Revised: Results – Meta-Analysis
"In the Bayesian model, between-study heterogeneity was estimated as a posterior median τ^2 of 3.3 (95% CrI: 1.0 to 9.5), corresponding to an I^2 of 84.7% (95% CrI: 63.9% to 94.2%)."

Reviewer 1, comment 5
In Section 3.3, the authors state: "In the context of meta-analysis, shrinkage is comparable to the inverse-variance weighting in the frequentist framework." This statement is potentially misleading. Bayesian shrinkage of study-specific effects occurs because posterior estimates incorporate information from the hierarchical distribution and borrow information from the overall mean. In contrast, inverse-variance weighting in frequentist meta-analysis determines the contribution of each study to the pooled effect estimate but does not change the individual study estimate. The authors should clarify whether they are discussing the estimation of individual study effects or the weighting mechanism used to obtain the pooled effect.

Response
We thank the reviewer and agree that this statement might be misleading and have revised the text to clarify that shrinkage affects individual study estimates, while inverse-variance weighting affects each study's contribution to the pooled estimate.

Revised: Results – Meta-Analysis
"The single study estimates shown in Fig. 2 differ between frameworks because Bayesian shrinkage pulls individual study estimates toward the pooled mean, borrowing information from the overall mean. This has no equivalent in the frequentist framework, where the contribution of each study to the pooled effect is determined by inverse-variance weighting. The degree of shrinkage does, however, depend on each study's precision in a way that parallels inverse-variance weighting: more precise studies are shrunk less (and are also weighted more heavily toward the frequentist pooled estimate), while less precise studies are shrunk more (and weighted less)."

Reviewer 1, comment 6

In Section 3.4, the authors conclude: "This shows that the results are robust to prior specification." This conclusion should be restricted to the specific prior distributions examined in this study. The results demonstrate robustness under the selected scenarios, but do not imply that Bayesian results are generally insensitive to prior choices.

Response

We thank the reviewer for raising this and have adjusted the wording to clarify that the results show robustness to the alternative priors tested in the sensitivity analysis (see revisions below). Following the reviewers second comment, we have added two additional prior sensitivity analyses and would like to gently refer the reviewer to our reply to their second comment to see the changes made in the Results – Sensitivity Analysis section.

Revised: Methods – Meta-Analysis – Sensitivity Analysis

"To examine the sensitivity of the results to the choice of prior distributions, we fit the Bayesian model again for alternative sets of priors that we consider to be extreme but still reasonable."

Reviewer 1, comment 7

In the Methods section and Discussion, the authors state that all code and data are publicly available on GitHub. However, the provided GitHub link is currently inaccessible. Since reproducibility is an important component of this study, the authors should provide a valid and accessible repository link before publication.

Response

We thank the reviewer for raising this. The GitHub repository is now publicly accessible at github.com/PaulinaSell/lead-IQ_meta-analysis.

Reviewer 1, comment 8

In the Discussion, the statement that the Bayesian model yielded lower pooled effects "due to posterior shrinkage" is incomplete. The difference between Bayesian and frequentist estimates also reflects the incorporation of prior information and should not be attributed solely to shrinkage.

Response

We thank the reviewer and agree and have added that the difference in pooled estimates is also attributable to the incorporation of prior information.

Revised: Discussion

"This is attributable to both the prior for the overall effect (μ) and the posterior shrinkage

of individual study estimates, influenced by the heterogeneity prior (τ), as illustrated by the sensitivity analyses (Table 4)"

Minor comments

Reviewer 1, comment 9
In Supplementary File 5, Figures 5a and 5b require clearer descriptions. The figure legends should specify what is displayed and how the prior predictive and posterior predictive checks were conducted.

Response
We thank the reviewer and added figure descriptions to Supplementary File 5.

Reviewer 1, comment 10
In Section 3.4, the manuscript states that sensitivity analysis results are presented in Table 4, but they appear to be reported in Table 5. Please correct this reference.

Response
We thank the reviewer and we have corrected the table numbering and referencing.

Reviewer 2

This manuscript compares Bayesian and frequentist random-effects meta-analysis using the association between blood lead levels and children's IQ as a case study. The topic is relevant, and the use of the same studies and effect estimates in both analyses makes the comparison easy to follow. However, several methodological and reporting issues should be addressed before publication

Major comments

Reviewer 2, comment 1
The manuscript compares one REML model with one Bayesian normal-normal hierarchical model. This is a useful applied example, but it does not support broad conclusions about Bayesian versus frequentist frameworks. With only 12 studies and substantial heterogeneity, the frequentist analysis could also include a small-sample adjustment. The authors should either broaden the comparison or describe the study more cautiously as a comparison of two specific implementations.

Response
We thank the reviewer for this comment and fully acknowledge that a single application cannot provide a definitive comparison between both statistical frameworks, and have therefore weakened conclusions that are beyond the scope of our applied study, and have added two sentences to the limitation section in our Discussion, cautioning against interpreting our results too generally (see revisions below).

In doing so, we have focused our comparison on general aspects, such as the role of shrinkage, that we can observe in our analysis through, for example, the sensitivity analyses we have added. In addition to our pre-existing description of how shrinkage operates in our specific sample, we now better explain which study settings our findings are expected to generalize to (small sample, high heterogeneity), and where shrinkage-induced differences in the pooled estimate are likely to be smaller (large sample, low heterogeneity). We described these analyses in more detail in our reply to reviewer 1's comment 2.

Furthermore, we thank the reviewer for pointing out that the sample size in our study warrants a small-sample adjustment. We have replaced the unadjusted analysis with an analysis using the Knapp-Hartung adjustment; the confidence interval of the pooled estimate in the adjusted analyses are now ~0.2 wider. The adjustment is briefly introduced in the Method section (see revisions below).

Revised: Title
"Quantifying uncertainty: A practical comparison of Bayesian and frequentist meta-analytic frameworks applied to lead exposure and children's IQ"

Revised: Abstract – Background
We performed frequentist and Bayesian meta-analyses to compare uncertainty quantification and applicability for risk assessment using lead exposure and children's IQ as a case study.

Revised: Abstract – Results
"Depending on the specified prior configuration, the Bayesian model was less sensitive to studies reporting substantial uncertainty, resulting in a lower pooled effect compared to the frequentist model. This regularization is particularly valuable in small meta-analyses, where frequentist heterogeneity estimates can be unstable."

Revised: Method – Meta-Analysis – Frequentist Meta-Analysis
"For the frequentist meta-analysis, we employed a random-effects model with a small-sample adjustment, [...]"

Revised: Results – Meta-Analysis
"Additional file 6 contains the full model code and output for both models. The results of both models are shown as forest plots in Figure 2."

Revised: Discussion
"By applying a parallel approach to both meta-analytic models — including the use of identical effect estimates, a multilevel random-effects model, and I^2 as a common measure of heterogeneity — we provide a direct comparison of a frequentist and a Bayesian random-effects meta-analysis."

"Whilst our study illustrates some of the differences of both Bayesian and frequentist approaches in the context of the here analyzed data, it should not be taken as a comprehensive comparison of both statistical frameworks. Such a comparison would require setting up a simulation study to systematically modify and quantify differences, which was outside the scope of the present study."

Revised: Conclusion
"This study provides both an updated evidence synthesis of epidemiological studies examining the association between lead exposure and children's IQ, and an applied methodological comparison of the Bayesian and conventional frequentist frameworks for meta-analysis."
"In the context of our study, comparing both meta-analytic models showed that the Bayesian approach offered distinct advantages for risk assessment and policy-making by fully quantifying uncertainty."

Reviewer 2, comment 2
Although the coefficients were converted to the same logarithmic scale, the included studies differ considerably in age, exposure range, study design, IQ instrument, and covariate adjustment. Some estimates also come from models adjusted for other environmental exposures. Therefore, the coefficients may not represent exactly the same estimand. Authors need more caution explaining those numbers.

Response
We thank the reviewer for this comment. We agree and have added a more explicit statement to the Discussion, emphasizing that the coefficients pooled in our study may not represent exactly the same estimand which complicates the interpretability of the pooled estimate.

Revised: Discussion
"The included studies were heterogeneous with respect to covariate adjustment, exposure range, IQ measurement instrument and cohort characteristics, meaning that the coefficients pooled in this study may not represent exactly the same estimand. This limits the generalizability of the pooled estimate and complicates its interpretation for specific populations."

Reviewer 2, comment 3
The manuscript states that study estimates in both panels of Figure 2 are shrunk toward the pooled estimate. However, the Bayesian panel appears to show posterior study-specific estimates, whereas the frequentist panel shows the original study estimates. Inverse-variance weighting of the pooled mean is not the same as shrinkage of study-specific effects. The statement that frequentist methods implicitly use flat priors should also be removed or qualified. Similarly, the claim that the Bayesian model is less sensitive to outliers is too strong because the model still assumes normally distributed random effects and no formal outlier analysis is presented.

Response
We thank the reviewer for pointing out the misleading terminology relating to Figure 2. We have adjusted the wording (see below, "Results").

Furthermore, we agree that in the context of our manuscript it is neither necessary nor helpful to give frequentist methods a Bayesian interpretation. We have replaced it with a more nuanced description of the differences in the two meta-analytic frameworks we wanted to highlight here (see below, "Discussion").

Regarding the reviewers third point, we acknowledge that the claims we make regarding outlier treatment in the Bayesian model are too strong, and thank them for this comment; we believe that with its help, we could significantly improve the clarity and argumentation in our manuscript. Since reviewer 1 has raised a similar point, we omit repeating our reply and the associated changes here, and would instead gently refer the reviewer to our reply to reviewer 1´s comment 2.

Revised: Results
"The single study estimates shown in Fig. 2 differ between frameworks because Bayesian shrinkage pulls individual study estimates toward the pooled mean, borrowing information from the overall mean."
"This has no equivalent in the frequentist framework, where the contribution of each study to the pooled effect is determined by inverse-variance weighting."

Revised: Discussion
"In contrast, frequentist models rely on inverse-variance weighting that is fully determined by the data – uncertain studies are weighed less, but the degree to which this happens cannot be configured by the researcher. Setting the model up for stronger shrinkage via priors can therefore be particularly relevant in meta-analyses analysing only a small set of studies."

Minor comments

Reviewer 2, comment 4
Different values appear in the Abstract, Results, Figure 2, and Table 4. Check the consistency.

Response
We thank the reviewer for pointing out these numerical inconsistencies. We have fixed these issues in the Abstract, Results, Figure 2 and Table 4.

Reviewer 2, comment 5
The text refers to children aged 0–11 years, while Table 1 includes participants under 18, pregnant women, prenatal exposure, and methylmercury.

Response

We thank the reviewer for raising this — the prose description did not fully match the PECOTS criteria in Table 1. We removed it and refer readers directly to Table 1, which is now the only source for this information.

Revised: Methods – Literature Review

"Publications were considered relevant according to predefined PECOTS – Population, Exposure, Comparator, Outcome, Timing of exposure and Setting (Table 1)."

Reviewer 2, comment 6

Check the confidence interval for frequentist heterogeneity estimate. For examples, the reported value tau2 is 3.3 with 95% CI [1, 3.3]. Seems not usual that the point estimate is equal to the upper limit. Need to double check this.

Response

We thank the reviewer for catching this. The upper limit indeed was wrong: a coding error caused the point estimate of τ^2 to be printed in place of the upper confidence limit. We have corrected this, and the confidence interval is now reported from the correct bound returned by `metafor`.

Revised: Results – Meta-Analysis

"The frequentist model estimated a closely comparable τ^2 *= 3.3 (95% CI: 1.0 to 11.7) […]"*